

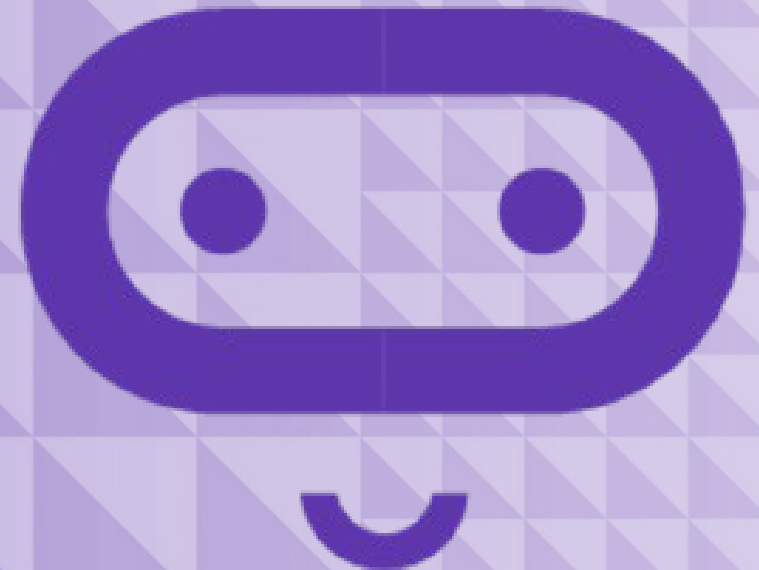
micro:bit

Introduction to Computer Science

Unit 7: Coordinates

Lesson A

Understanding coordinates



What we'll learn in Lesson A

- The micro:bit coordinate grid and how it can be used

Coordinates in math

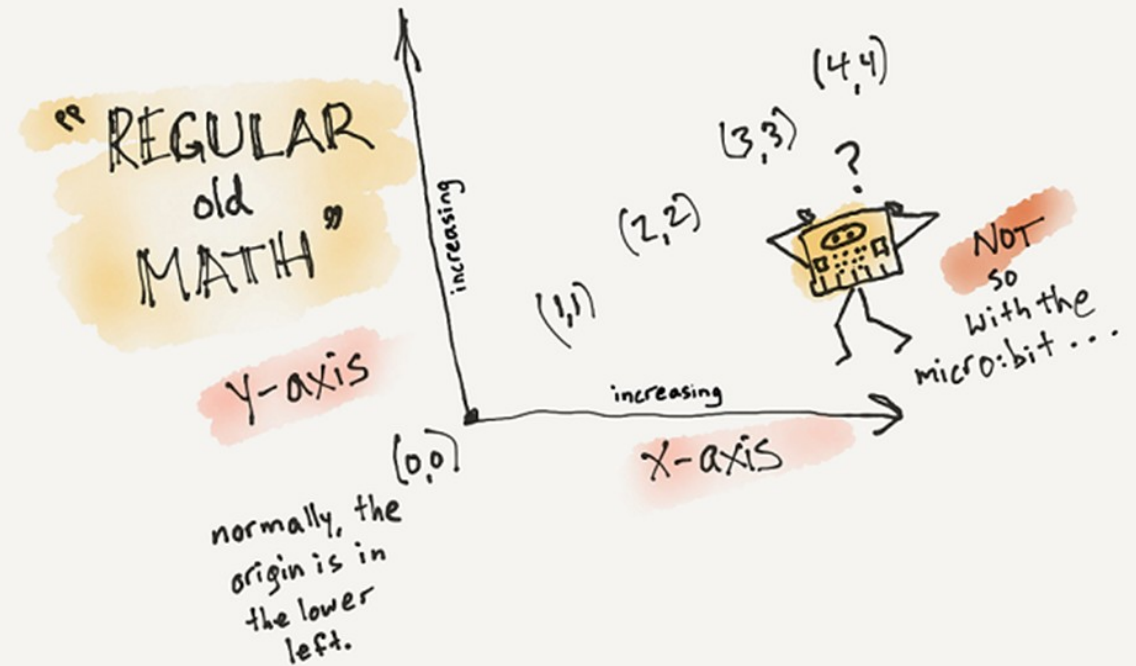
Axes

The basic coordinate grid has two axes:

- **X-axis:** runs horizontally (left to right)
- **Y-axis:** runs vertically (up to down)
- **Origin**

The two axes meet at a point called the **origin**, where both the x- and the y-values are **zero**.

On this basic coordinate grid, the origin is in the lower left corner of the grid and has the coordinates **(0,0)**.



Coordinates in math (cont.)

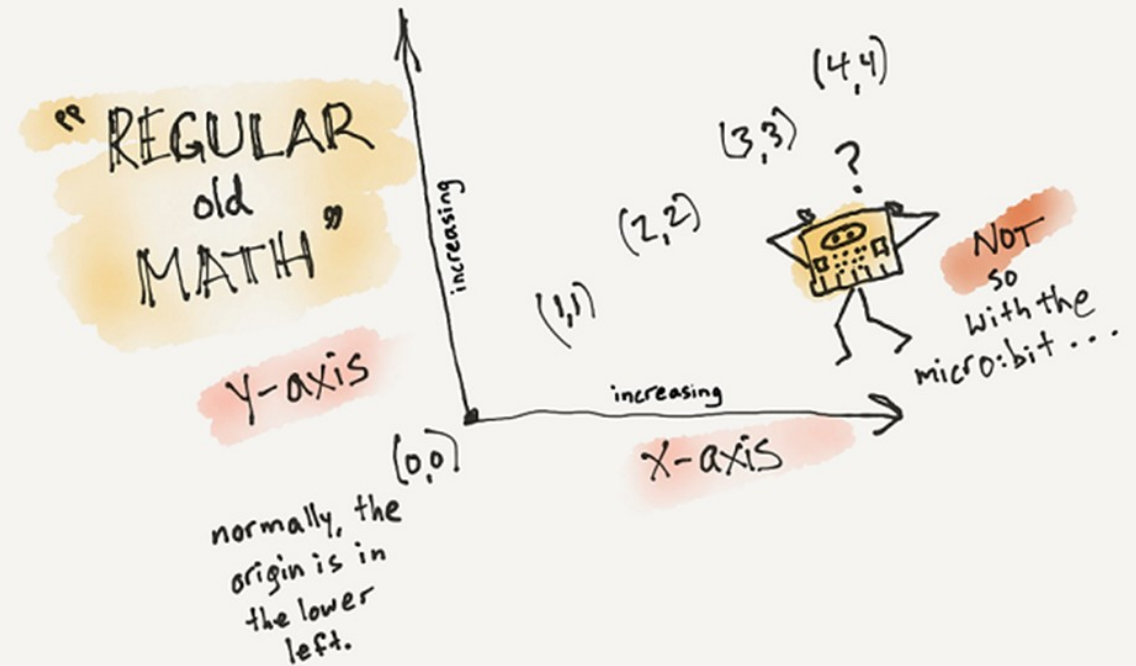
Coordinate pair

The **first value** in a coordinate pair is the **x-value** and the **second value** in a coordinate pair is the **y-value**: (x, y)

Coordinate value changes

The value of the **x-coordinate** increases from **left to right** and is a measure of how many units a point is **horizontally** from the origin.

The value of the **y-coordinate** increases from **bottom to top** and is a measure of how many units a point is **vertically** from the origin.



Coordinates in micro:bit

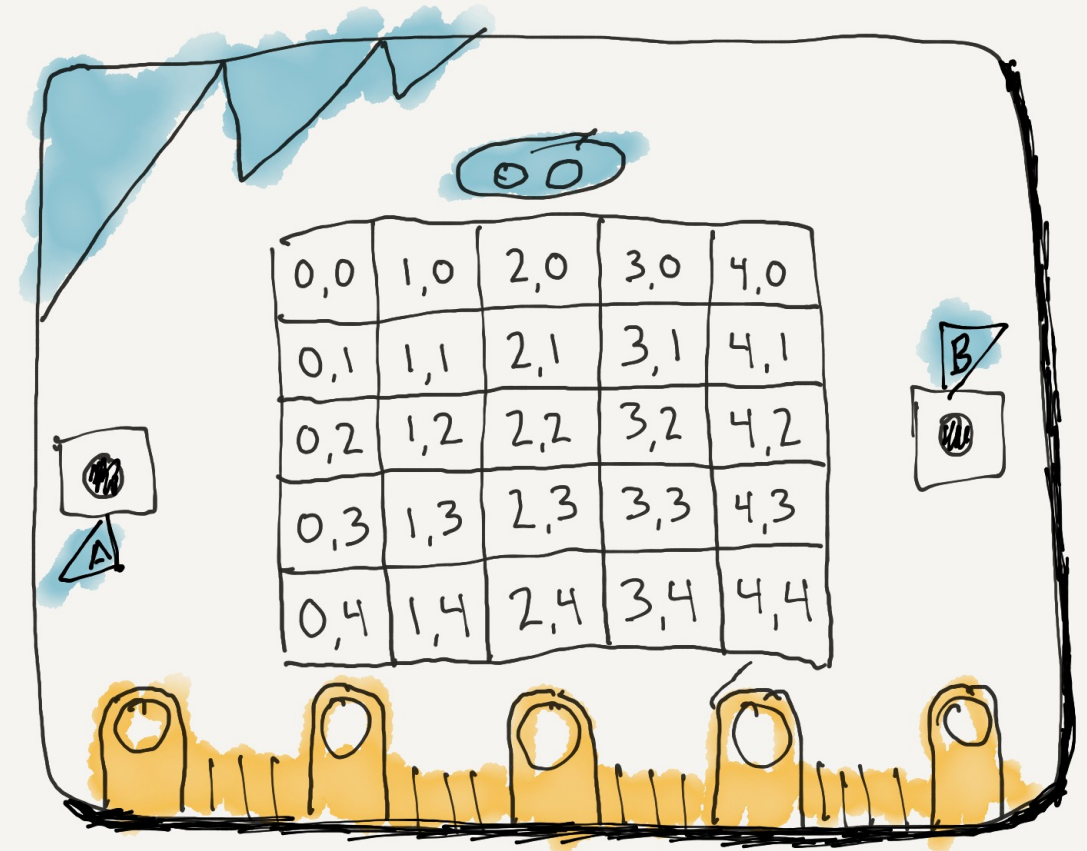
Coordinate grid on the micro:bit

The 5 x 5 grid of LEDs on the micro:bit represent a coordinate grid with a horizontal x-axis and a vertical y-axis.

It has an origin and you can refer to the position of the LEDs with coordinate pairs.

There are similarities and differences between the micro:bit coordinate grid and basic coordinate grids in math class.

MICRO:BIT LED
grid (x,y) coordinates



Coordinates in micro:bit are...

Similar to the coordinate grid in math:

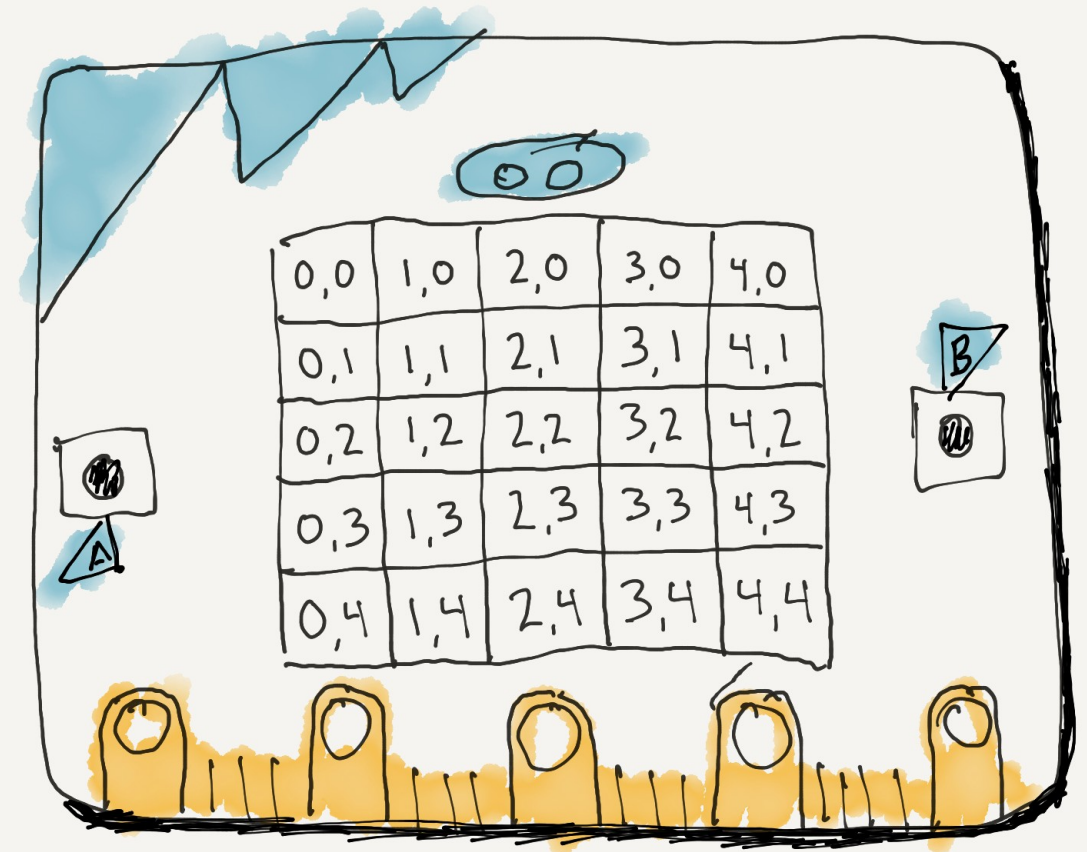
The values of the **x-coordinates** range from 0 through 4 and increase from **left to right** just as they do in the coordinate grids used in math.

Different from coordinate grids in math:

The **origin (0,0)** is in the **top left corner**.

The values of the **y-coordinates** range from 0 through 4 and increase from **top to bottom**.

MICRO:BIT LED
grid (x,y) coordinates



René Descartes (1596–1650)

French philosopher and mathematician who developed the coordinate system we use today

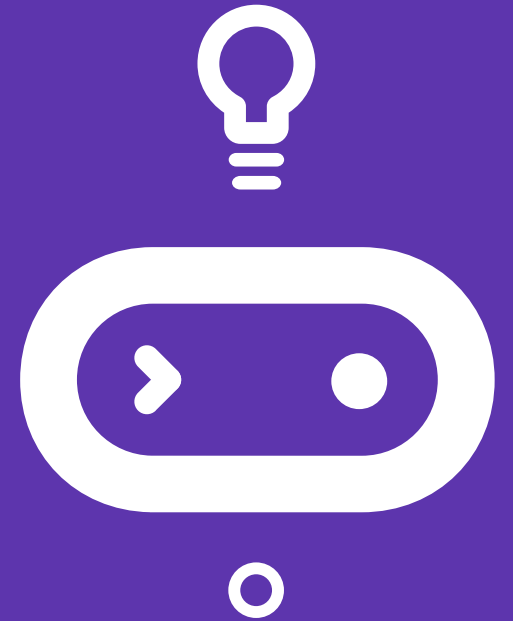
A story goes that while lying in bed, he noticed a fly on the ceiling. In wondering how he could describe the fly's exact position on the ceiling, he decided to use a corner of the ceiling as a reference point and then describe the fly's position as a measure of how far away from the reference point one would need to travel horizontally and then vertically to reach the fly.

His coordinate system proved useful in many ways including creating an important link between the studies of algebra and geometry. Geometric shapes could now be described by points on a coordinate plane.



Let's review!

1. What location is the **origin** on the micro:bit screen?
2. What is the range of x-values on the micro:bit screen and in which direction do they increase?
3. What is the range of y-values on the micro:bit screen and in which direction do they increase?



Battleship

In pairs, play the game Battleship using a 5 x 5 coordinate grid drawn on two pieces of paper from (0, 0) to (4, 4) just like the micro:bit screen.

- Plot where your ships are on your grid.
- Players call out shots using (x, y) coordinate pairs.

Player's Grid

0,0	B	B	B	B
D	1,1	2,1	3,1	4,1
D	1,2	2,2	3,2	4,2
D	1,3	2,3	3,3	4,3
0,4	1,4	2,4	P	P

Battleship
BBBB

Destroyer
DDD

PT Boat
PP

Opponent's Grid

0,0	1,0	2,0	3,0	4,0
0,1	X	2,1	3,1	4,1
0,2	1,2	2,2	3,2	X
0,3	1,3	2,3	X	4,3
0,4	1,4	2,4	X	4,4

X = miss

■ = hit

What we learned in Lesson A

- The micro:bit coordinate grid and how it can be used



Next time: Lesson B

- Code animations and patterns
- Assess our learning with a quiz



Lesson B

Coding animations and patterns



What we'll learn in Lesson B

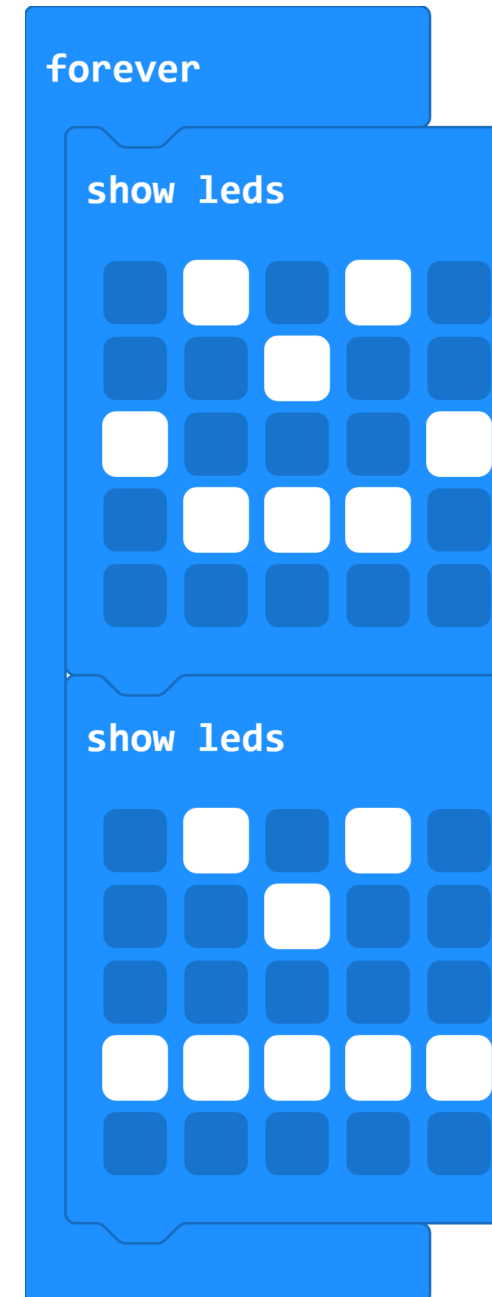
- Use coordinates to create animations and patterns
- Review how much we've already learned

Smile animation

1. Create a smiling and non-smiling face with 'show leds' blocks.
2. Recreate this program using the LED Plot coordinate blocks.

Hints:

- What LEDs change between the two faces?
- Use the LED Toggle block for the LEDs that change.

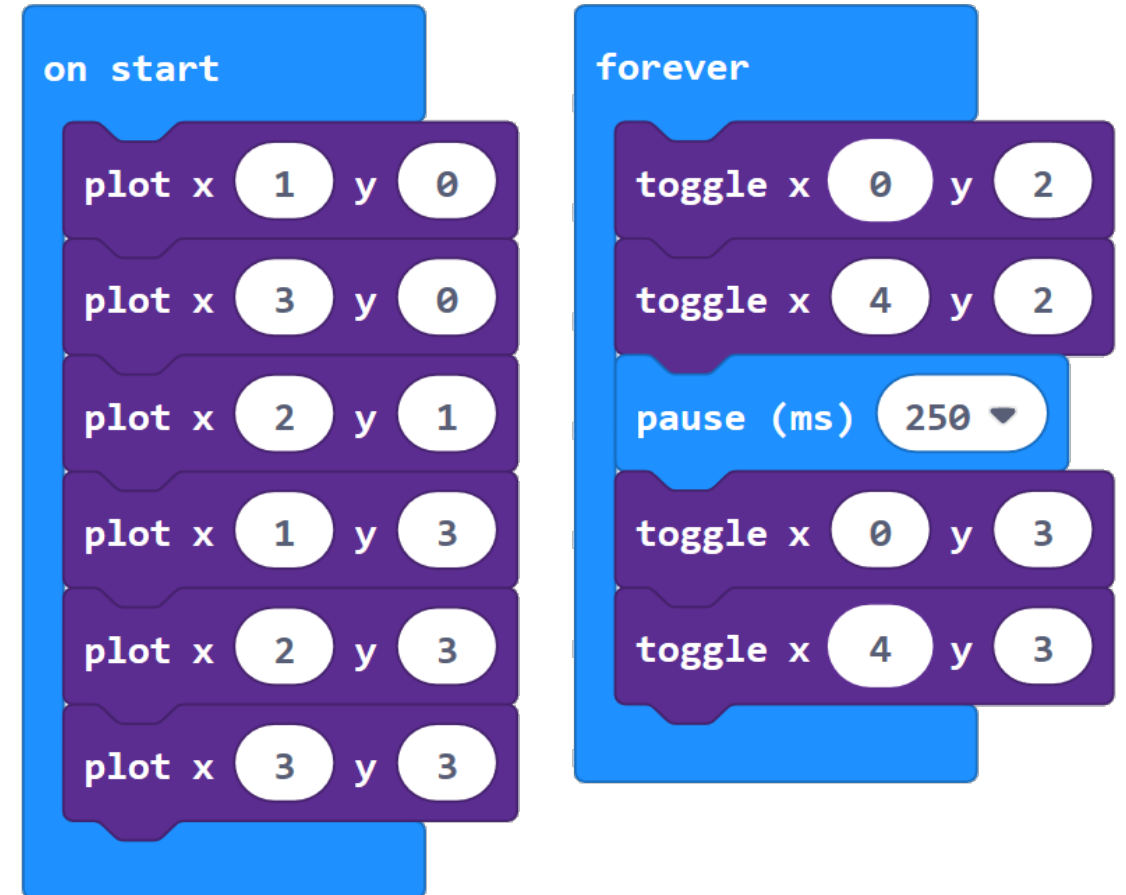


Smile animation code

Your program might look like:

Mods:

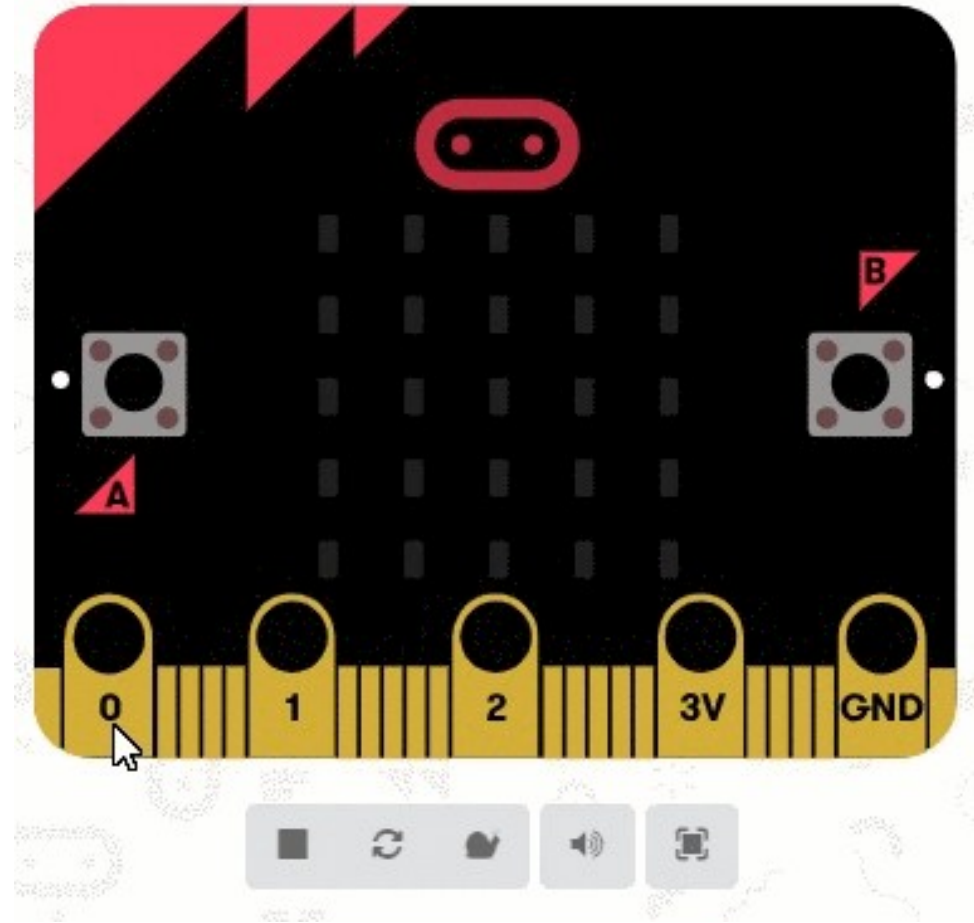
- Add a third image to the animation, perhaps a frown face.
- Make your own custom animation! What LEDs stay the same and which need to be toggled?



Random pattern generator

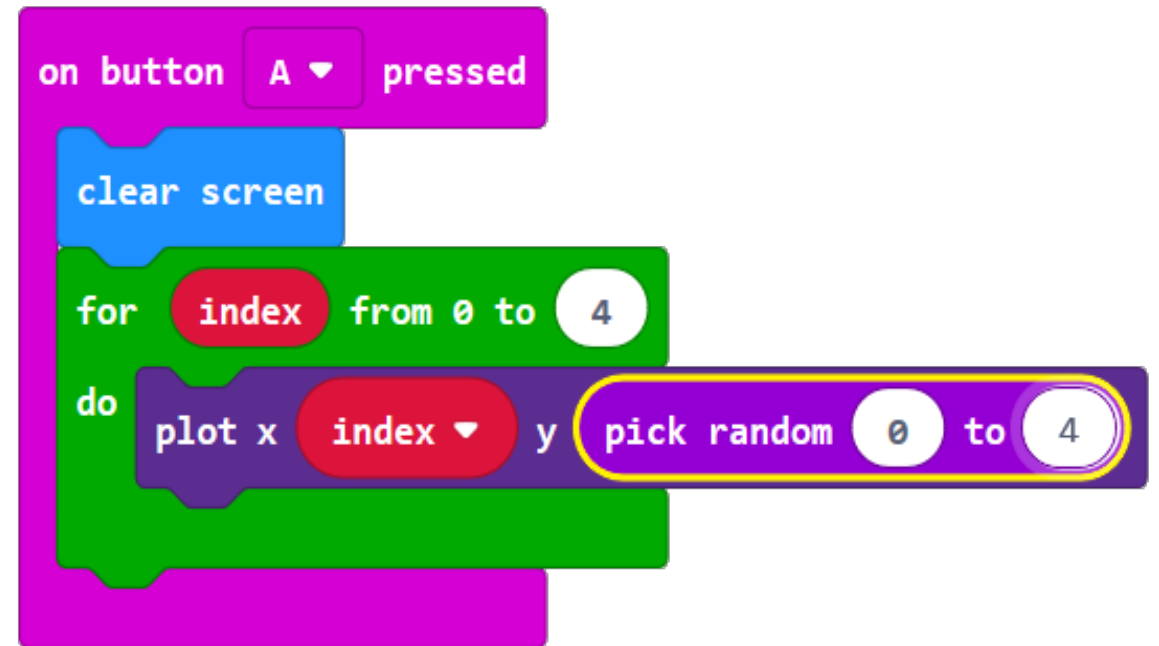
Pseudocode:

- On button A pressed, use a loop to turn on a random set of LED lights on the micro:bit.
- Your display will have one LED lit for each column (or x-coordinate value) from 0 through 4.



Random pattern generator: Code button A

- Clear the screen
- Use a for loop
- Plot an LED
- Use the for loop index as the x-value
- Use a random number from 0 to 4 as the y-value



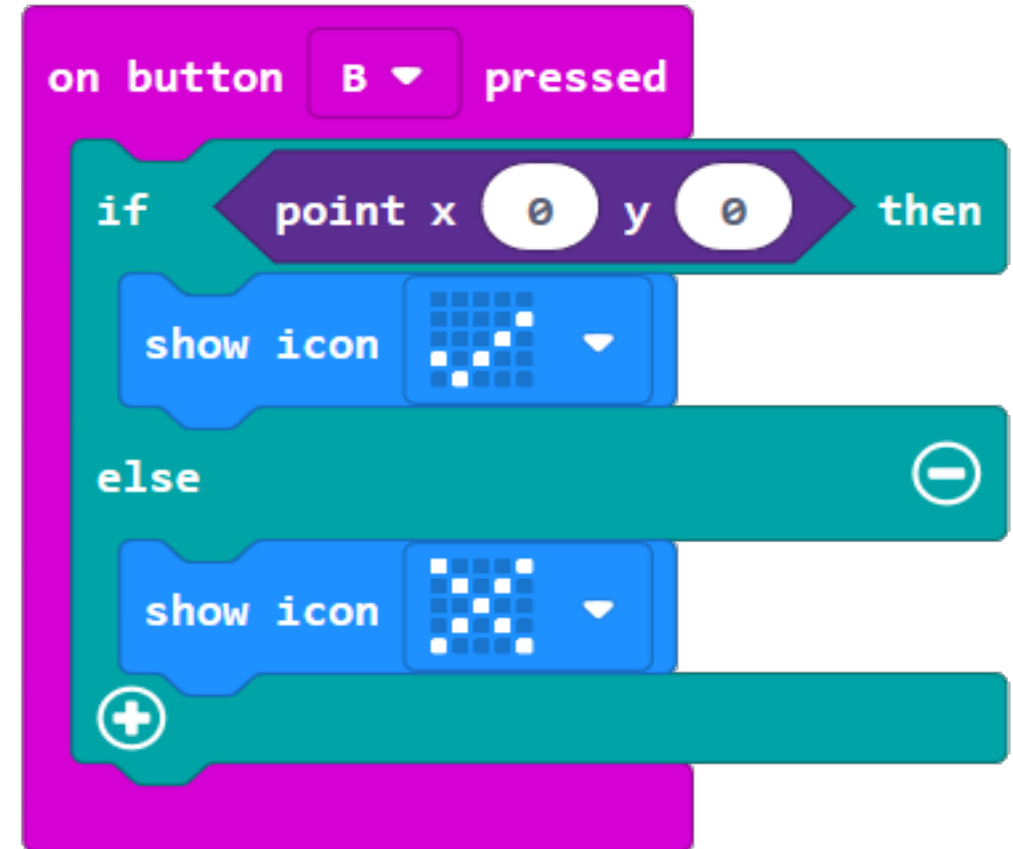
Random pattern generator: Code button B

Check whether a certain LED is lit

- Use an **IF... THEN... ELSE...** **block** to check whether LED Point (0, 0) is on.
- If so, display a yes (check) mark.
- Else, display a no (X) mark.

Mod:

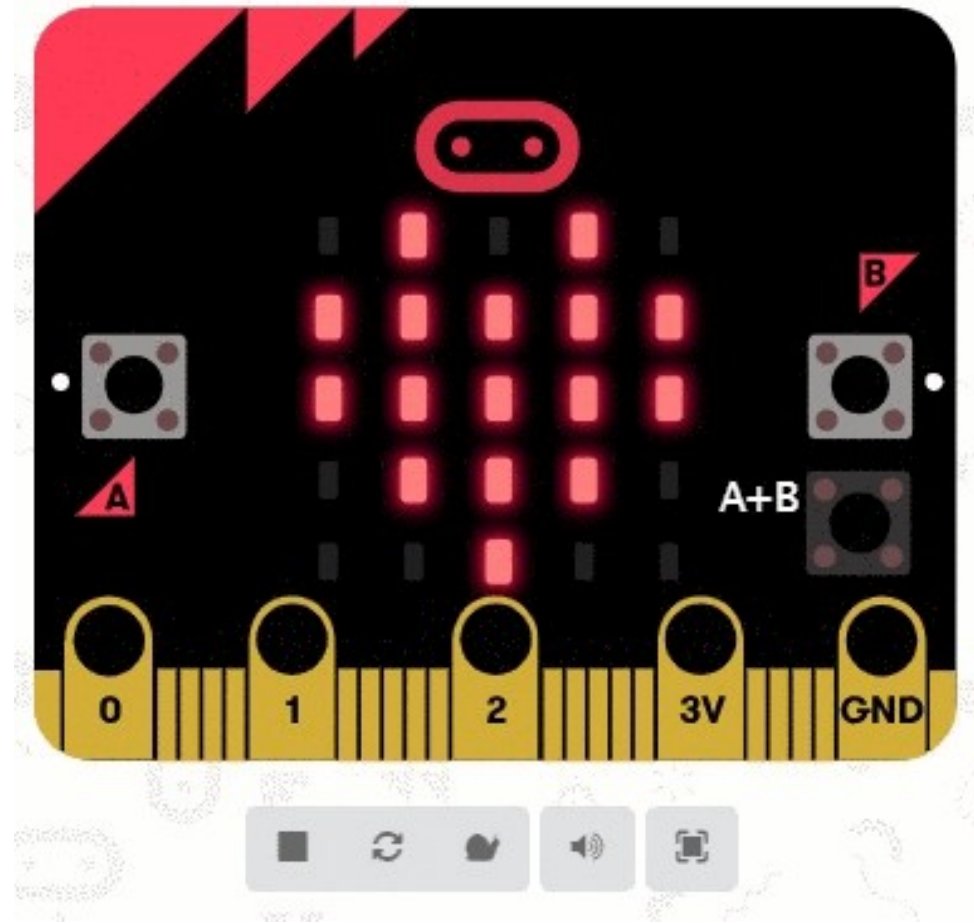
- Add a loop to test for all coordinates on the micro:bit when button B is pressed.



Brightness

Pseudocode:

- Set the brightness level for the LEDs to the highest level on start and use button A to decrease the brightness, and button B to increase the brightness.
- Use on button A+B pressed to check and display numerically the current brightness level.



Brightness settings

On start

- Set brightness = 255
- Show an icon

Button A:

- Reduce brightness by 25

Button B:

- Increase brightness by 25

Button A + B together:

- Show the brightness number

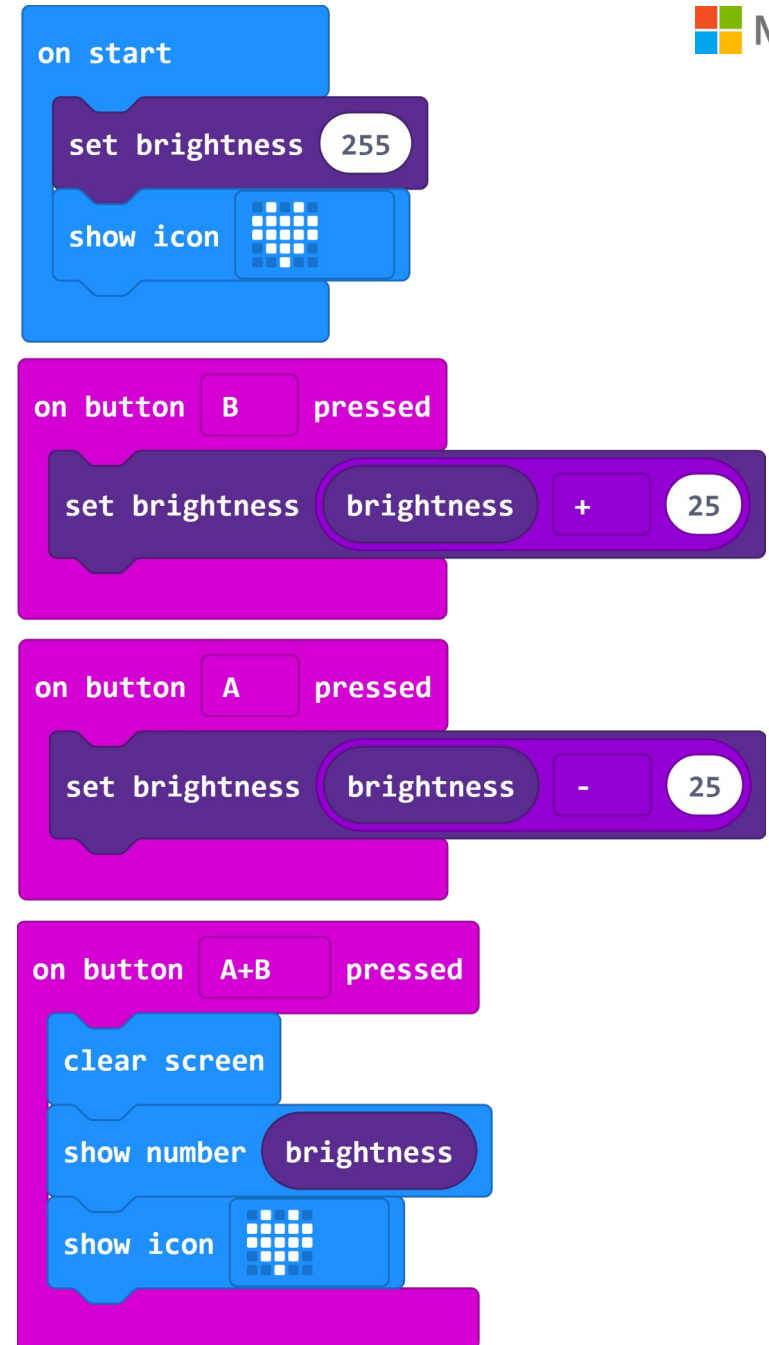


Brightness code

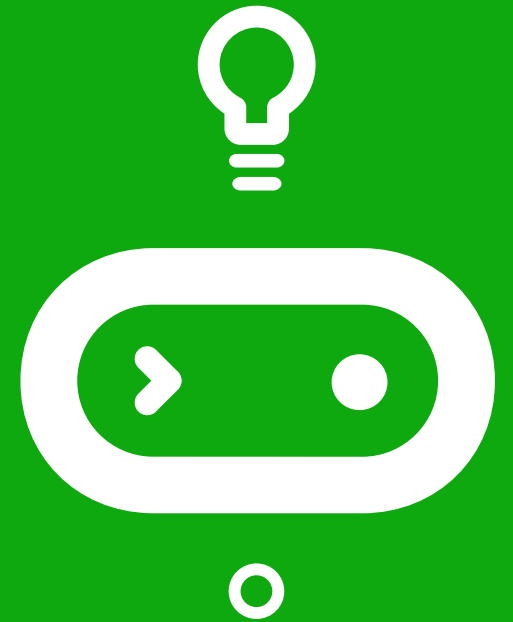
Your program might look like:

Mod:

What happens if adding 25, or subtracting 25, from the current brightness level would result in a sum or difference outside of the 0 to 255 brightness range? Add code to protect that from happening.



Time for a quiz!



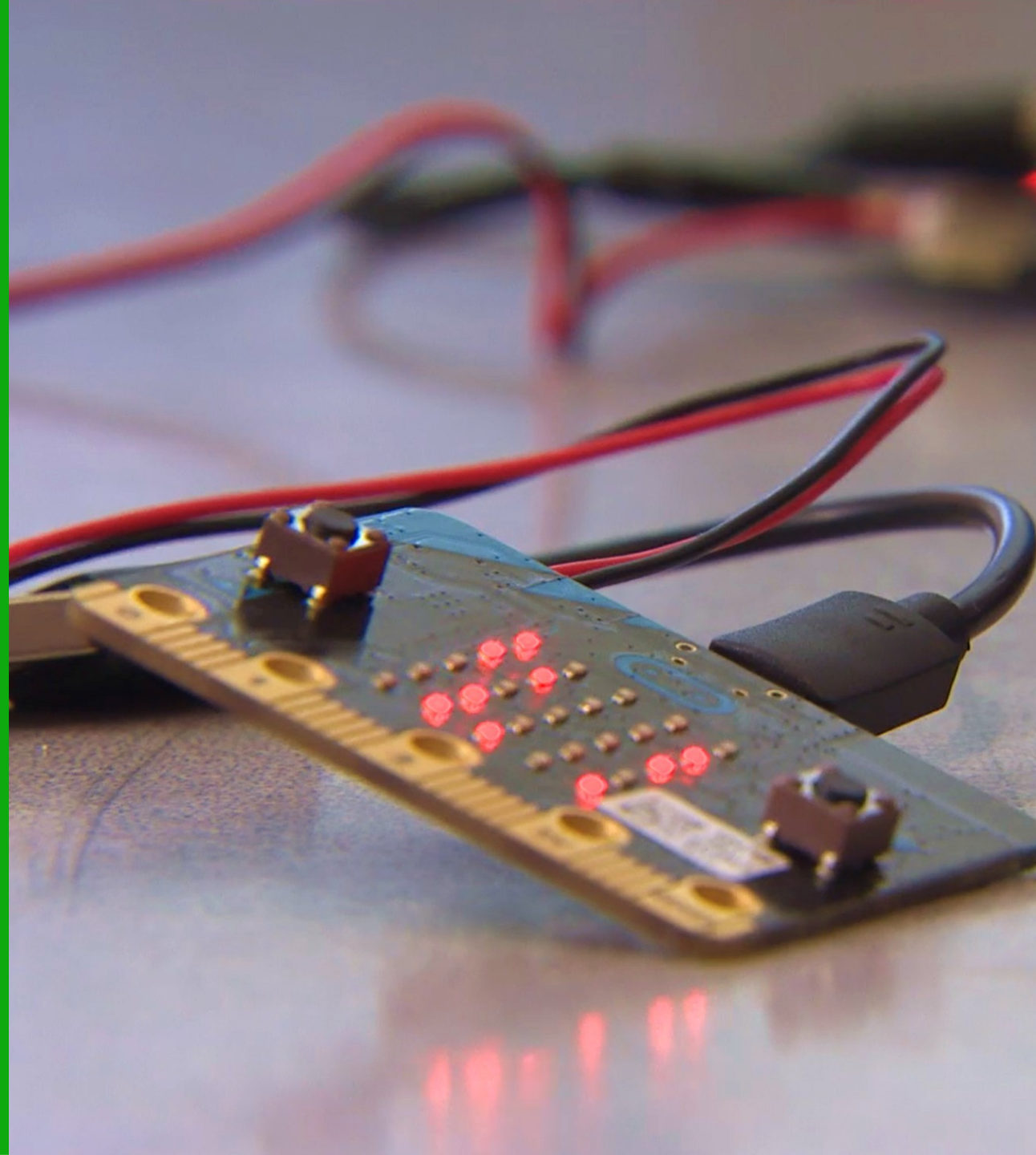
What we learned in Lesson B

- Used coordinates to create animations and patterns
- How much we've already learned!



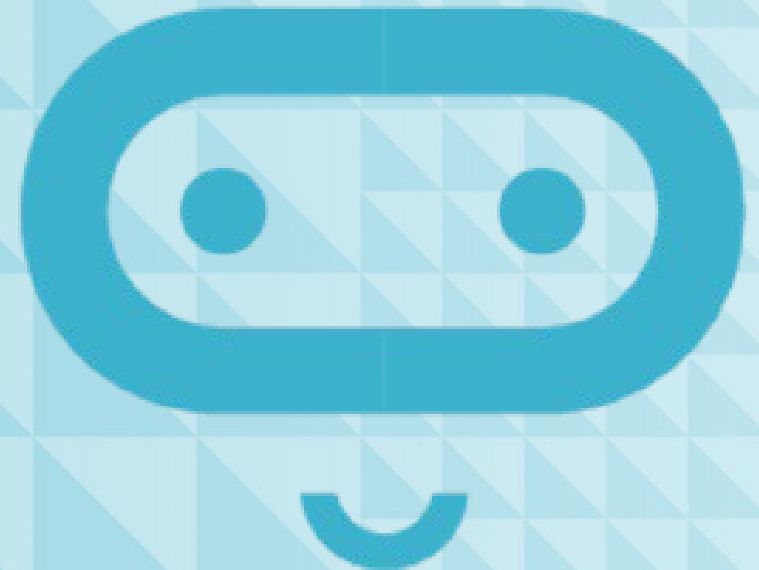
Next time: Lesson C

- Screensaver or game project



Lesson C

Screensaver or game project



What we'll learn in Lesson C

- Code a micro:bit program that uses coordinates to make a screensaver or a game

Screensaver or project?

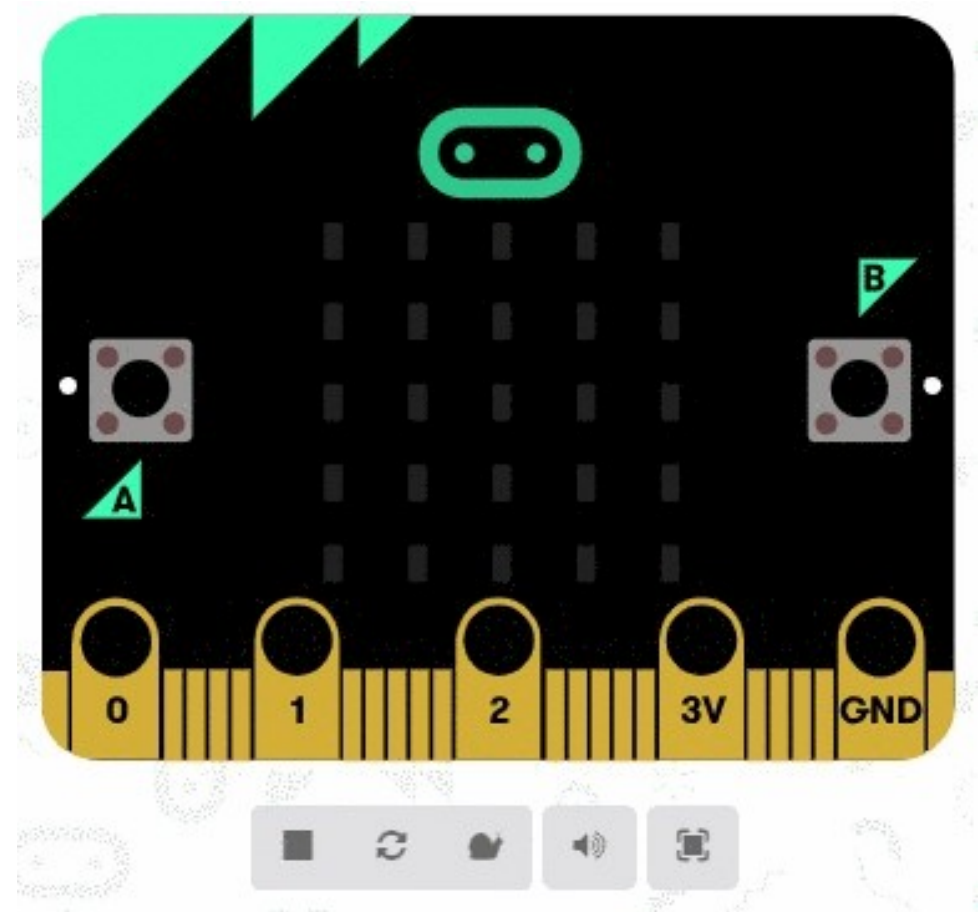
Use what you now know about LEDs, coordinates, and brightness to create your own screensaver, or game.

Create:

- A “screensaver” animation using the plot/unplot blocks. You can fill the screen line by line, pausing between each one, or fill it with a random constellation of stars.

Or

- A game that uses sprites to manage the x- and y-coordinate values of the different objects.



Some project ideas

Fireworks screensaver

This project uses a for loop with the plot/unplot blocks to create a symmetrical design on the screen. The sample code uses a subtraction operation to get a variable that decreases as the index variable in the loop increases.

Sample code: [makecode.microbit.org/ fJsPc3Jr3THk](https://makecode.microbit.org/fJsPc3Jr3THk)

Cascade screensaver

This example creates a diagonal cascading effect across the screen. Note the use of a variable (speed) to allow you to easily change the speed of the animation by changing just one number value. Try modding a current board game.

Sample code: [makecode.microbit.org/ _1Jg5KJDg1AHr](https://makecode.microbit.org/_1Jg5KJDg1AHr)

Dodge ball game

This is a Dodge Ball game that uses one sprite (dodger) to try to avoid another sprite (ball). You use the A and B buttons to move the dodger to avoid the balls that are falling from the top of the screen.

Sample code: [makecode.microbit.org/ UvU5eaEgg3xw](https://makecode.microbit.org/UvU5eaEgg3xw)

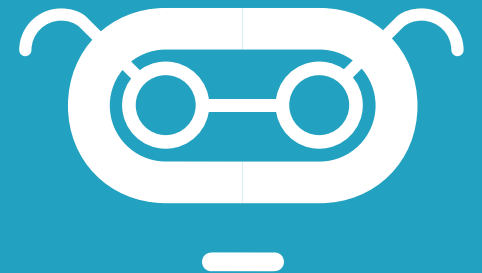
Project expectations

Use a design thinking approach and ensure the project meets these specifications:

- Uses at least three of the different kinds of 'plot', 'unplot', 'toggle', 'point x y' blocks, and uses variables to update the coordinates in some way
- Uses plotted LEDs in a meaningful way
- The program compiles and runs as intended and includes meaningful comments in code
- Provide the written Reflection Diary entry

Reflection Diary

- Did you create a screensaver? A game? Something different? How did you decide?
- If you created a game, what is the object of the game?
- How does your project use coordinates?
- Describe something in your project that you are proud of.
- Describe a difficult point in the process of designing this program and explain how you resolved it.
- What feedback did your testers give you?
How did that help you improve your design?
- Publish your MakeCode program and include the link.



What we learned in Lesson C

- Coded a micro:bit program that uses coordinates to make a screensaver or a game

